Homework 3

Released November 2026 • Practice for Test 3 Instructors: Steven Wu, Giulia Fanti

Q1. Mean estimation: central versus local (8 points)

Each of $n = 2{,}500$ users holds a value $x_i \in [0, 4]$, and we want the mean $\mu = \frac{1}{n}\sum_i x_i$. Neighbouring datasets differ in one user's value. The *error* of an estimate $\tilde{\mu}$ is its expected absolute error $\mathbb{E}[|\tilde{\mu} - \mu|]$, over the randomness of the algorithm. You may use two facts: if $Z \sim \mathrm{Lap}(b)$ then $\mathbb{E}[|Z|] = b$, and for any random variable W, $\mathbb{E}[|W|] \le \sqrt{\mathbb{E}[W^2]}$.

(a) (2 points) A trusted curator releases μ plus Laplace noise with $\varepsilon = 1$. What are the sensitivity of μ and the noise scale, and what is the (expected) error?

(b) (2 points) With no trusted curator, user i sends one bit Y_i with $\mathbb{P}[Y_i = 1 \mid x_i] = \frac{1}{4} + \frac{x_i}{8}$. Show that this is ε-LDP and find the smallest ε.

(c) (2 points)

(i) Give an *unbiased* estimator $\hat{\mu}$ of μ computed from $\bar{Y} = \frac{1}{n}\sum_i Y_i$, meaning $\mathbb{E}[\hat{\mu}] = \mu$.

(ii) Show that $\mathrm{Var}[\hat{\mu}] \le \frac{16}{n}$, and use this to bound the expected error $\mathbb{E}[|\hat{\mu} - \mu|]$ for $n = 2{,}500$.

Hints: a bit $Y \in \{0, 1\}$ has $\mathrm{Var}[Y] \le \frac{1}{4}$; for independent variables the variances add; $\mathrm{Var}[cW] = c^2 \mathrm{Var}[W]$; and since $\hat{\mu}$ is unbiased, $\mathbb{E}\left[(\hat{\mu} - \mu)^2\right] = \mathrm{Var}[\hat{\mu}]$.

(d) (1 point) Ignoring constants, how does the ratio (local error)/(central error) depend on n, and why?

(e) (1 point) In the local protocol of parts (b)–(c), every user sends their bit Y_i to an untrusted curator, who then publishes only the estimate $\hat{\mu}$. Viewing the whole pipeline $\mathbf{x} = (x_1, \ldots, x_{2500}) \mapsto \hat{\mu}$ as a mechanism, what (central) differential privacy guarantee does it satisfy, and why is it not $2{,}500 \ln 3$?

Q2. True or false (6 points)

For each of the following statements, say whether it is true or false **and explain your answer**. An unexplained verdict earns no credit.

(a) (2 points) *In PATE, the released noisy labels can only be differentially private if each teacher model was itself trained with a differentially private algorithm.*

(b) (2 points) *Three hospitals compute the total of their patient counts using secure summation, so the server learns nothing about any single hospital's count.*

(c) (2 points) *Alice encrypts two different messages m_1, m_2 of the same length with the same uniformly random one-time-pad key k: $c_1 = m_1 \oplus k$ and $c_2 = m_2 \oplus k$. An eavesdropper who sees both c_1 and c_2 learns nothing about m_1 and m_2, because each ciphertext on its own is uniformly random.*

Q3. Subsample-and-aggregate and PATE (6 points)

A company has 2,000 user records. It splits them, using a random permutation fixed before looking at the data, into $k = 50$ disjoint blocks of 40 records. Neighbouring datasets differ in one record.

(a) (2 points) On each block j, an arbitrary (possibly non-private) algorithm produces a number $z_j \in [0, 1]$. The company releases the average of the z_j plus Laplace noise with $\varepsilon = 0.25$. What are the sensitivity of the average and the noise scale, and why is the release DP even though the block algorithms are not private?

(b) (1 point) For PATE on this data, the votes of the 50 teachers over four classes form a histogram. What is its ℓ_1-sensitivity, and what Laplace scale per count makes one released label ε_q-DP?

(c) (1 point) The system answers 25 queries, each at $\varepsilon_q = 0.2$. What is the total guarantee of the released labels and of a student model trained on them?

(d) (2 points) For each change, say whether the guarantee from part (c) still holds.

(i) Assign records to teachers by hashing a public record ID, fixed in advance.

(ii) The next query point is chosen by looking at the noisy labels released so far.

(iii) The next query point is chosen by looking at the *exact*, un-noised vote counts of the previous query.

(iv) Sort the records by income and give the first 40 to teacher 1, the next 40 to teacher 2, and so on.

Q4. Privacy auditing (4 points)

An auditor is given a mechanism that is claimed to be $(\varepsilon, \delta) = (1, 0.1)$-DP. She picks two neighbouring datasets $\mathbf{x}, \mathbf{x}'$: $\mathbf{x}$ *contains* the target record, and $\mathbf{x}'$ is the same dataset with the target record replaced by a different record. She runs an attack T that outputs 1 for "the target record is in the dataset", and measures TPR $= \mathbb{P}[T(A(\mathbf{x})) = 1]$ and FPR $= \mathbb{P}[T(A(\mathbf{x}')) = 1]$.

(a) (2 points) One attack achieves TPR $= 0.5$ and FPR $= 0.1$. Does it contradict the claim? What lower bound on ε does it imply when $\delta = 0.1$?

(b) (1 point) A second attack gets TPR $= 0.2$ and FPR $= 0.1$. Is $\varepsilon = 1$ consistent with this attack, and does it show the mechanism is $(1, 0.1)$-DP?

(c) (1 point) The auditor cannot compute TPR and FPR exactly. She runs the mechanism 1,000 times on each dataset and records the observed fractions, for example 100 false alarms in 1,000 runs, an observed FPR of 0.10. Because the runs are random, the true probability may differ a little from the observed fraction. A 95% *upper confidence bound* U for a probability p is a number computed from the runs such that, with probability at least 95% over the randomness of the runs, $p \le U$ (for 100 false alarms in 1,000 runs, roughly $U \approx 0.12$). A 95% *lower confidence bound* L similarly guarantees $p \ge L$ with probability at least 95%. Why should the auditor plug a lower confidence bound for TPR and an upper confidence bound for FPR into $\ln \frac{\text{TPR}-\delta}{\text{FPR}}$, rather than the observed fractions?

Q5. The local model, shuffling and federated learning: true or false (5 points)

For each statement, say whether it is true or false **and explain your answer**. An unexplained verdict earns no credit.

(a) (1 point) *To estimate a mean to error α with ε-differential privacy, a trusted central curator needs roughly $1/(\alpha\varepsilon)$ users (ignoring constants), while the local model needs roughly $1/(\alpha^2\varepsilon^2)$ users: quadratically more.*

(b) (1 point) *In federated SGD each client sends only a gradient, never its raw data, so the server cannot learn anything about a client's training examples.*

(c) (1 point) *Only two clients take part in a round, and the server sees just the sum of their updates through secure aggregation. Even if client 2 colludes with the server, client 1's update stays hidden.*

(d) (1 point) *In each round of federated learning, every client k sends the server an update Δ_k (the change to the model computed on its own data). The server* aggregates *these into one number per coordinate, the sum $\sum_k \Delta_k$, and uses it to update the model. To protect users, the server adds Gaussian noise to this sum before using it. Two federated datasets are* user-level neighbours *if they differ in the entire data of one client. Then adding Gaussian noise to the sum gives user-level differential privacy, even if the updates Δ_k are* not clipped *to any maximum norm.*

(e) (1 point) *In a federated learning system, each client clips its update and adds noise to it* on its own device, *so every noisy update is the output of an ε_0-local-DP randomizer. The noisy updates are sent to a* shuffler, *which removes the clients' identities and forwards the updates to the server in random order. Normally this hides each update among all the others and gives the server's view a stronger central guarantee. Now suppose the shuffler is compromised and tells the server which update came from which client. Each client is still protected by ε_0-local DP.*

Q6. ElGamal encryption (4 points)

All arithmetic is modulo 11. Bob uses ElGamal with base $g = 2$: secret key $x = 4$ and public key $y = g^x = 5$. To encrypt m with fresh randomness k, Alice sends $(r, c) = (g^k,\ m \cdot y^k)$.

(a) (2 points) Alice encrypts $m = 3$ with $k = 2$. Compute (r, c) and show how Bob decrypts. *Useful facts:* $5^2 \equiv 3$, $4^2 \equiv 5$, $3 \cdot 4 \equiv 1$.

(b) (2 points) The ciphertext of $m' = 2$ with $k' = 3$ is $(8, 8)$. The server multiplies $(4, 9) \cdot (8, 8)$ entry by entry. What does Bob get on decrypting? Explain in one sentence why this works. *Useful fact:* $10^2 \equiv 1$.

Q7. Auditing AI reasoning (4 points)

Below are two transcripts in which an AI assistant was asked a question about security or privacy and reasoned step by step to an answer. For each, say whether the final answer is **correct** or **incorrect**. If incorrect, give the **number of the first step that is wrong** and say what is wrong with it. Either may be correct.

(a) (2 points)

Prompt. n users each apply randomized response with $\varepsilon_0 = \ln 3$ to a private bit. A shuffler permutes the n reports uniformly at random and publishes the shuffled list. Without using any amplification-by-shuffling theorem, what central privacy guarantee can we state for the published list?

Assistant's reasoning.

1. Each user's report is $\ln 3$-LDP.
2. Local privacy implies central privacy for the list of reports: neighbouring datasets differ in one bit, so only one report changes distribution, and the likelihood ratio of any output list is at most 3.
3. The shuffler applies a random permutation that does not depend on the data, so the published list is post-processing of the list of reports.
4. Post-processing cannot increase privacy loss, so the shuffled list is $\ln 3$-DP.
5. So we can state $\varepsilon = \ln 3$. Shuffling can only help beyond that.

(b) (2 points)

Prompt. A company trains a model by federated learning. Clients send updates through secure aggregation, so the server only ever sees the sum of the clients' updates. The model is then released to every user. Does secure aggregation alone protect clients against membership inference on the released model?

Assistant's reasoning.

1. Secure aggregation hides each client's individual update, and the server learns only the sum.
2. The new model is computed from that sum and then released.
3. The released model is therefore a function of the sum of the updates, which depend on all clients' data.
4. Since the server never sees any single client's update, the released model does not leak information about any individual client.
5. So membership inference on the released model is not a concern. Yes, it protects clients.

Q8. Local histograms

(4 points)

Each user holds a category $x_i \in \{1, 2, 3, 4\}$ and sends a vector of four bits $b = (b_1, \ldots, b_4)$, where $b_j = \mathbb{I}[x_i = j]$ and then *each bit is flipped independently with probability* $1/4$.

(a) (2 points) What is the smallest ε for which this randomizer is ε-LDP? Explain why it is not $4 \ln 3$.

(b) (1 point) Let h_j be the true fraction of users in category j and $\bar{b}_j$ the average of the received jth bits. Give an unbiased estimator of h_j.

(c) (1 point) We want the whole report to be $\ln 3$-LDP instead. What flip probability should each coordinate use? (Leave surds in your answer.)

Optional. Beaver triple

Not in exam scope

This problem is optional and is **not in scope for Test 3**; it concerns multiplying secret-shared numbers, which we only touched on.

Work modulo 11. Parties 1 and 2 hold shares of $x = 3$ as $(5, 9)$ and of $y = 4$ as $(7, 8)$. A dealer gives shares of a triple $a = 2, b = 5, c = ab = 10$: Party 1 holds $(a_1, b_1, c_1) = (1, 3, 6)$, Party 2 holds $(1, 2, 4)$. Compute $d = x - a$ and $e = y - b$ by having the parties open their shares, then find each party's share $z_i = c_i + d\,b_i + e\,a_i$ (Party 1 also adds de) and check that the shares add to xy.